

Yujin Kim

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Engineering Experiences:

Georgia Science and Engineering Fair (GSEF)

spring 2026

- 2nd in State (Material Science)
- Researched the application of Shear Thickening Fluid (STF) to 3D-printed Thermoplastic Polyurethane (TPU) lattice structures to evaluate their mechanical performance and energy absorption
- Collected shear and compression data on the Universal Testing Machine to analyze the greatest resistance to deformation ratio

Governor's Honors Program (GHP) Civil & Environmental Engineering

summer 2026

- Selected as 1 of 6 statewide finalists
- Proposed a tunnel design with quality control measures aligned with Codes of Records for seismic and shear strain calculation
- Created Piping and Instrumentation (P&ID) diagrams and created molds for concrete pour using woodworking tools

VEX Robotics

2022- 2025

- Qualified for Georgia VEX State/World Championship Competition
- Worked on programming robot on Computer-Aided Design (CAD) and documented team contributions on logbook
- Led outreach for the GSMST VEX organization, social media, and sponsorship efforts

Savannah College of Arts and Design (SCAD) Summer Seminars

summer 2025

- Full scholarship recipient; completed Industrial Design & Architecture workshops
- Designed a key holder in Onshape CAD and 3D-printed a functional prototype to address the problem of misplaced campus lanyards. Integrated a motion sensor and LED light to provide visual illumination when movement is detected

Technology Student Association (TSA)

2024 - Present

- Ranked 6th Place National | Manufacturing Prototype/ Engineering CAD
 - Developed a 3D printed puzzle with an integrated keyhole based on a year-long design prompt; modeled the prototype in Onshape and fabricated it using 3D printing.
- Ranked 4th Place National | Structural Design
 - Designed and constructed competition bridges from balsa wood, applying structural design principles to meet annual competition specifications.

Kia Motors Manufacturing Georgia/SAE International's A World In Motion

Spring

- Built a paper car powered by airflow, competing in three on-site competition categories: accuracy, distance, and speed
- 1st place in distance and accuracy

Academic Studios- Co-founder

2024 - Present

- Create study resources for audience of 33,000+ students through social media; Google Classroom community with 250+ members

Art Experiences:

Scholastic Art & Writing Awards

2024 & 2025

- 1- Gold Key | 2- Silver Key | 6- Honorable Mention

Georgia Art Education Association (GAEA) All-State Art finalist

Spring 2026

- 1 of top 50 student artists chosen from over 500 state-wide entries

Atlanta High School Art Exhibition (AHSAE) Dogwood

Spring 2026

- 1st placed (mixed media) \$300 prize money

- Lucy Award (Top Watercolor piece) \$100 prize money
- Piece title: Ethereal- Enhanced intricate patterns with white gel-pen to create contrast and depth, while manipulating alcohol ink drops with bent wires to create a dynamic sense of movement and direction

National Art Honor Society (NAHS)- President

2026 - Present

- Organized art activities for club meetings and led outside events (Depressed Cake Shop) to raise funds; Painted shelter murals to advocate for mental health.

Music Experiences

Georgia Music Educators Association (GMEA) All-State Orchestra

2021 - Present

- Statewide auditions to participate in annual three-day program and perform at GMEA All State concert

Emory Youth Symphony Orchestra (EYSO)

2023 - Present

- Pre-college honor ensemble for high school instrumentalists (2.5hrs/week)

American String Teachers Association (ASTA)

Spring 2024

- Performed viola at national strings conference

Awards

- **Young Scientist & Research Award from U.S. Air Force** *Spring 2026*
- **National Merit Recognition Silver Award- Council of Youth Anti-Drug (COYAD)** **2023 - Present**
- **Gold & Silver Key- Scholastic Art & Writing Awards** **2024 & 2025**
- **Qualification for the VEX Robotics Competition World Championship** *Spring 2023*
- **Design Award- VEX Robotics Competition** *Spring 2023*
- **1st Place (Accuracy & Distance)- Kia Motors Manufacturing Georgia/SAE International** *Fall 2024*
- **1st Place (Mixed Media)- Atlanta High School Art Exhibition (AHSAE)** *Spring 2026*
- **NOCTI Engineering Certification** *Fall 2025*
- **GSMST Academic Excellence Award** *Fall 2024*
- **Superior Rating- GMEA Solo and Ensemble Performance Evaluation** **2023 & 2024**
- **AP Capstone** **2025**